

Breast Cancer Classification Using Machine Learning

Comparative Analysis of Classification Algorithms

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Problem Statement

- Breast cancer diagnosis requires accurate classification of tumors.
- Manual diagnosis can be supported by machine learning models using diagnostic features.
- This project compares multiple classification algorithms to identify an effective predictive model.

Objective

01

Build a binary classification model to predict malignant vs. benign tumors

03

Train and compare multiple ML algorithms

02

Preprocess and prepare the dataset for machine learning

04

Select the best-performing model using classification metrics

Models Used

Logistic Regression

Linear classification using probability estimates

K-Nearest Neighbors

Distance-based classification using nearest neighbors

Decision Tree

Rule-based classification using hierarchical decision rules

Random Forest

Ensemble of decision trees for improved accuracy

Best Model: Logistic Regression

Performance Metrics:

Metric	Value
Accuracy	98.25%
Precision	98.61%
Recall	98.61%
F1 Score	98.61%
ROC-AUC	0.9954

Classification Report: Logistic Regression



The classification report details the performance of the Logistic Regression model for each class. For Class 0 (Benign), the model achieved a Precision of 0.98, Recall of 0.98, and an F1-score of 0.98. For Class 1 (Malignant), the model demonstrated slightly higher performance with a Precision of 0.99, Recall of 0.99, and an F1-score of 0.99. The Overall Accuracy of the model is 0.98.

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	98.25%	98.61%	98.61%	98.61%
KNN	95.61%	95.89%	97.22%	96.55%
Decision Tree	91.23%	95.59%	90.28%	92.86%
Random Forest	95.61%	95.89%	97.22%	96.55%

Best Model: Logistic Regression ✓

Methodology

Pipeline: Dataset → Data Preprocessing → Train/Test Split → Feature Scaling → Model Training → Model Evaluation → Best Model Selection

Algorithms Evaluated:

Logistic Regression

K-Nearest
Neighbors (KNN)

Decision Tree

Random Forest

ROC Curve & Model Evaluation

Logistic Regression ROC-AUC Score: 0.9954

The ROC curve demonstrates excellent discrimination between benign and malignant classes. An AUC of 0.9954 indicates near-perfect classification performance, with the model correctly distinguishing between tumor types across all classification thresholds.

Key Findings

Logistic Regression: Top Performer

Achieved the highest overall performance among all evaluated models.

Exceptional Class Discrimination

ROC-AUC of 0.9954 indicates exceptional class discrimination and model reliability.

High Accuracy and F1 Score

98.25% accuracy and 98.61% F1 score, demonstrating excellent classification capability.

Final Model Selection

Logistic Regression was selected as the final model for breast cancer tumor classification.

Conclusion

Machine learning can effectively classify breast cancer tumors using diagnostic features. Among the evaluated models, Logistic Regression provided the strongest overall performance with 98.25% accuracy and exceptional ROC-AUC of 0.9954.

Thank You